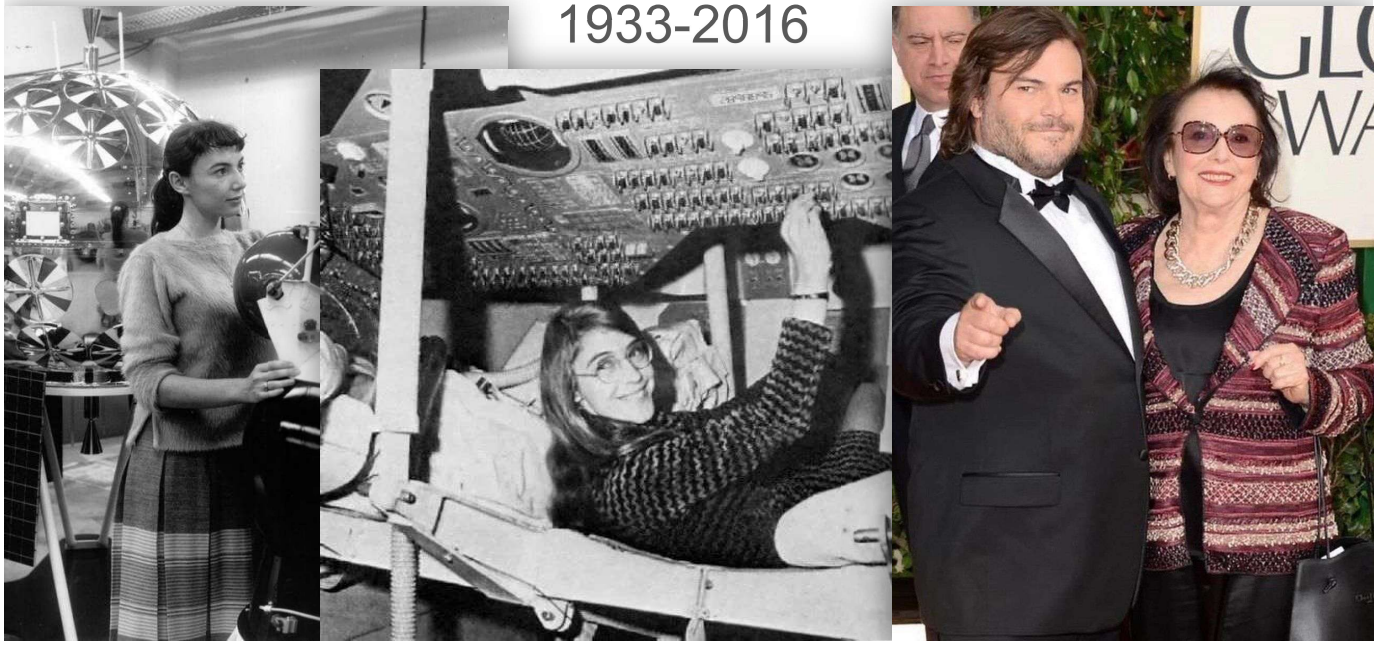
Judith Love Cohen

1933-2016



Taking Automation

To The Moon

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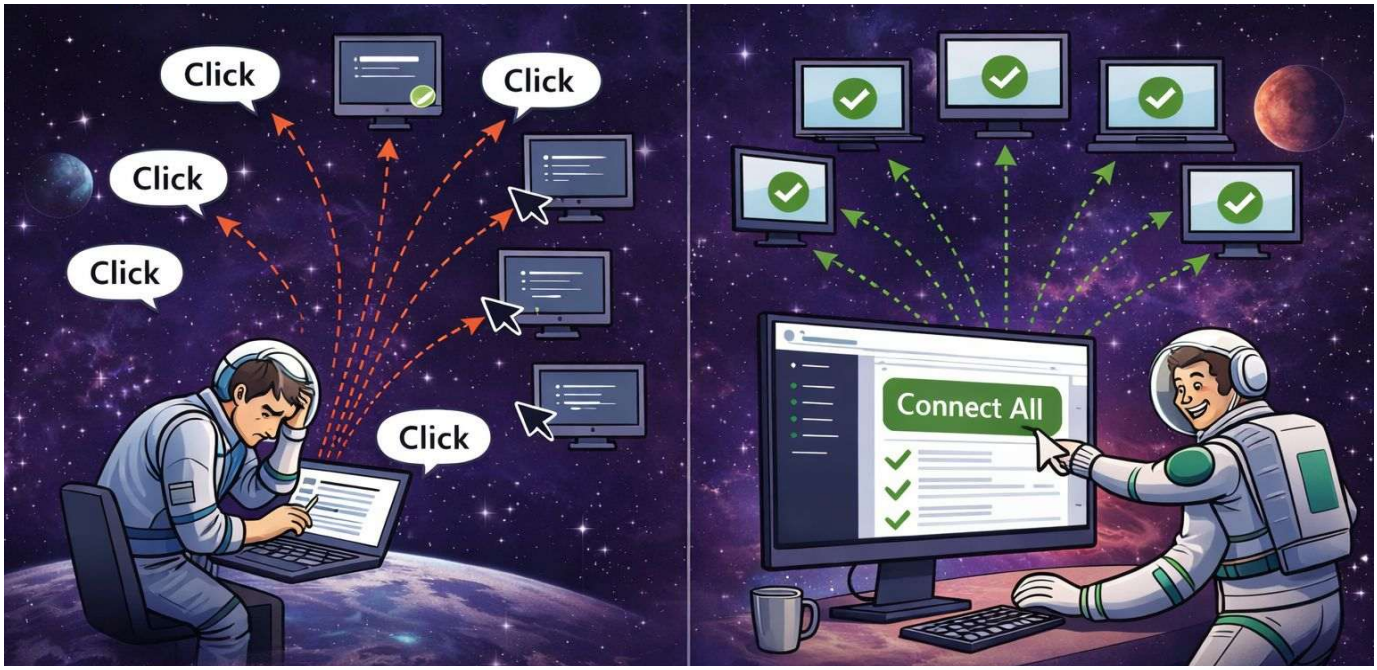
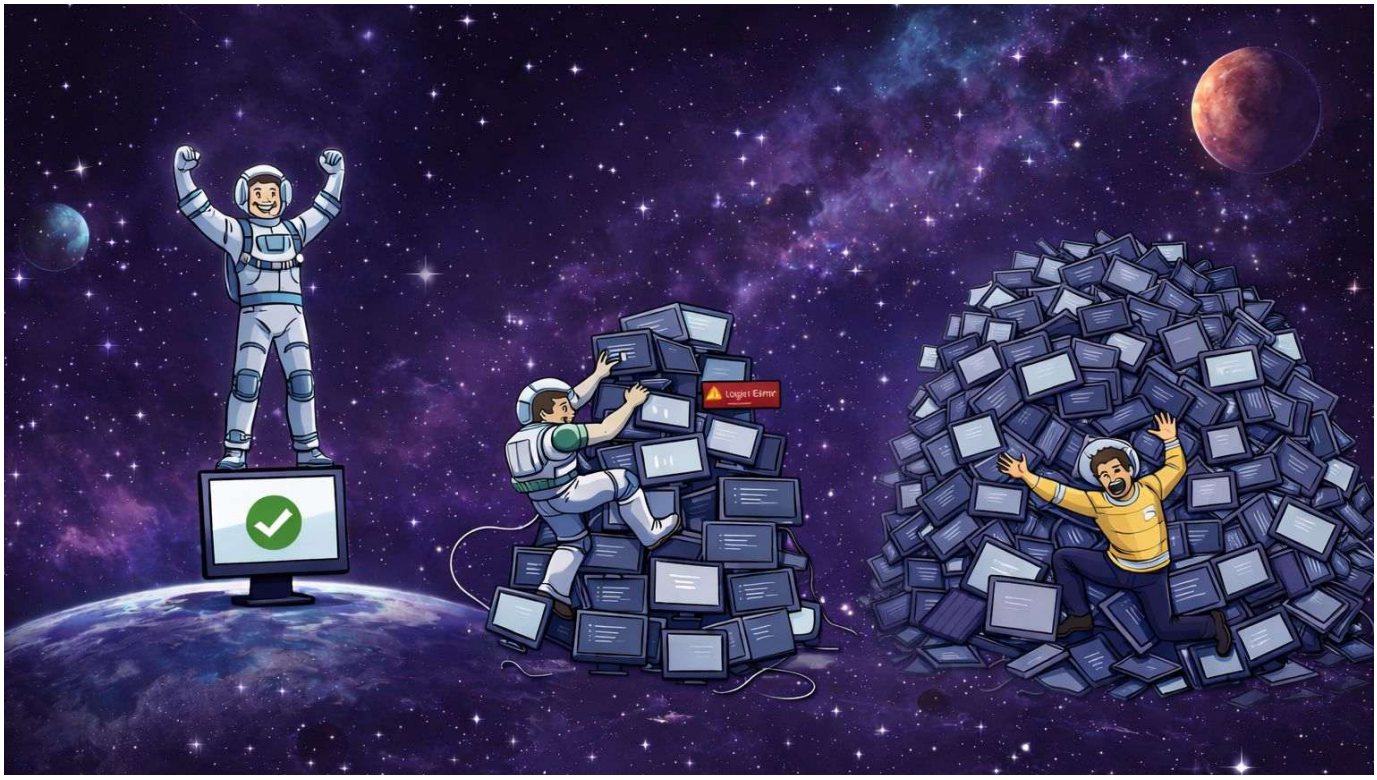
The Launch Problem





5. Mission Control:
Why PowerShell?

PowerShell was created by Microsoft to provide a powerful, consistent, and automatable way to manage Windows systems.

PowerShell can be used to interact with SMO.

SQL Server Management Objects (SMO) is a collection of objects that are designed for programming all aspects of managing Microsoft SQL Server.

Powershell modules dbatools & dbachecks connect PowerShell & SQL Server for you.

4. Powering Up:
Eliminating Manual Work

Examples:

- Automated backups across all instances
- Patch compliance checks across different versions of SQL Server
- Automated daily health checks

3. Weather Briefing:
Building Repeatable Infrastructure

Examples:

- Standardized server builds
- Scripted deployments
- Configuration drift detection
  Claudio Silva demo: https://youtu.be/AJP2JGHMlFk?si=0bYc1i9m6KUbEpW9&t=1005
- Disaster recovery validation automation

2. Area Evacuation:
Automation as Risk Reduction

Examples:

- Automated restore testing
- Alerting and reporting
- Proactive monitoring
- Self-healing scripts (where appropriate)

```
#Check SQL Patch levels for all instances
Test-DbaBuild -SqlInstance $sqlInstances -Update -MaxBehind 0CU
```

```
Build          : 15.0.2000
BuildLevel     : 15.0.2000
BuildTarget    : 15.0.4430
Compliant      : False
CULevel        :
CUTarget       : CU32
KBLevel        :
MatchType      : Exact
MaxBehind      : 0CU
NameLevel      : 2019
SPLevel        : RTM
SPTarget       : RTM
SqlInstance    : DESKTOP-RS3LEAE\DEMO2019
SupportedUntil : 1/8/2030 12:00:00 AM
Warning        :
```

```
#Run health checks on backups, last DBCC CHECKDB results
Invoke-DbcCheck -Check Backup, LastGoodCheckDb -SqlInstance $sqlInstances
```

```
      [+] Database SuperbowlPredictions has Data Purity Enabled on DBDIVA\DEMO 1ms
      [-] Database Pants last good integrity check should be less than 7 days old on DBDIVA\DEMO 2ms
        Expected the actual value to be greater than 2026-03-04T16:58:36.7057176Z, because You should have run a DBCC C
 heckDB inside that time, but got $null.
        221:                                $psitem.LastGoodCheckDb | Should -BeGreaterThan (Get-Date).ToUniversalTime
 ().AddDays( - ($maxdays)) -Because "You should have run a DBCC CheckDB inside that time"
        at <ScriptBlock>, C:\Program Files\WindowsPowerShell\Modules\dbachecks\3.0.2\checks\Database.Tests.ps1: line 22
 1
      [+] Database Pants has Data Purity Enabled on DBDIVA\DEMO 1ms
  Tests completed in 367ms
  Tests Passed: 14, Failed: 8, Skipped: 0, Pending: 0, Inconclusive: 0
```

```
#Install new instance of SQL Server with my config file
Install-DbaInstance -Version 2025 -SqlInstance DBDIVA\DEMO2025 -Path C:\SQL2025\StdDev_ENU
```

```
Copyright (c) Microsoft Corporation.

 Installing SQL Server (17.0) components on DBDIVA
    Installing SQL Server on DBDIVA from C:\SQL2025\StdDev_ENU\SETUP.EXE

PS C:\Presentations\DemoDBAVUG>
PS C:\Presentations\DemoDBAVUG> Install-DbaInstance -Version 2025 -SqlInstance DBDIVA\DEMO2025 -Path C:\SQL2025\StdDev_
ENU -NoPendingRenameCheck

Confirm
Are you sure you want to perform this action?
Performing the operation "Install 2025 from C:\SQL2025\StdDev_ENU\SETUP.EXE" on target "DBDIVA".
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "Y"): Y

ComputerName : DBDIVA
InstanceName : DEMO2025
Version      : 17.0
Port         :
Successful   : True
Restarted    : False
Installer    : C:\SQL2025\StdDev_ENU\SETUP.EXE
ExitCode     : 0
LogFile      : C:\Program Files\Microsoft SQL Server\170\Setup Bootstrap\Log\Summary.txt
Notes        : {}
```

```
<#
    More robust test of backups - Test-DbaLastBackup
    This dbatools command checks for a backup, tests the restore, AND runs DBCC checks.

    Don't run this in your production instances without understanding what it does!
    By default, it will run tests and restore copies of each of your databases on the SAME SERVER.

    https://dbatools.io/Test-DbaLastBackup/
#>
Test-DbaLastBackup -SqlInstance $sqlInstances
```

```
SourceServer   : localhost\DEMO2019
TestServer     : localhost\DEMO2019
Database       : Wikipedia
FileExists     : True
Size           : 2.84 MB
RestoreResult  : Success
DbccResult     : Success
RestoreStart   : 2026-03-11 13:05:04.926
RestoreEnd     : 2026-03-11 13:05:05.431
RestoreElapsed : 00:00:00
DbccMaxDop     : 0
DbccStart      : 2026-03-11 13:05:05.459
DbccEnd        : 2026-03-11 13:05:05.565
DbccElapsed    : 00:00:00
BackupDates    : {2026-03-11 11:56:59.000}
BackupFiles    : {C:\DemoBackups\DESKTOP-RS3LEAE$DEMO2019\Wikipedia\FULL\DESKTOP-RS3LEAE$DEMO2019_Wikipedia_FULL_20260
                 311_115659.bak}
```

```
#Export config files
Export-DbaInstance -SqlInstance $sqlInstances -Exclude Credentials, LinkedServers -Path $pathToExportInstance -Force
```

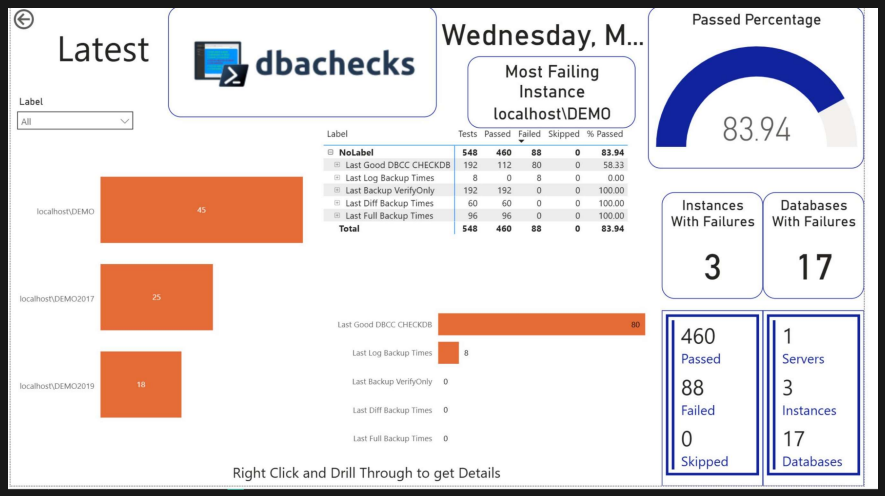
```
    Directory: C:\DbatoolsExport\localhost$DEMO2019


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         3/11/2026   1:10 PM           4175 sp_configure.sql
-a----         3/11/2026   1:10 PM            539 serverroles.sql
-a----         3/11/2026   1:10 PM            865 logins.sql
-a----         3/11/2026   1:10 PM           1364 dbmail.sql
-a----         3/11/2026   1:10 PM           3121 regserver.xml
-a----         3/11/2026   1:10 PM           5248 databases.sql
-a----         3/11/2026   1:10 PM              2 policymanagement.sql
-a----         3/11/2026   1:10 PM            837 resourcegov.sql
-a----         3/11/2026   1:10 PM          12834 extendedevents.sql
-a----         3/11/2026   1:10 PM          31446 sqlagent.sql
-a----         3/11/2026   1:10 PM            970 replication.sql
-a----         3/11/2026   1:10 PM         540778 userobjectsinsysdbs.sql
-a----         3/11/2026   1:10 PM           8151 oledbprovider.sql
```

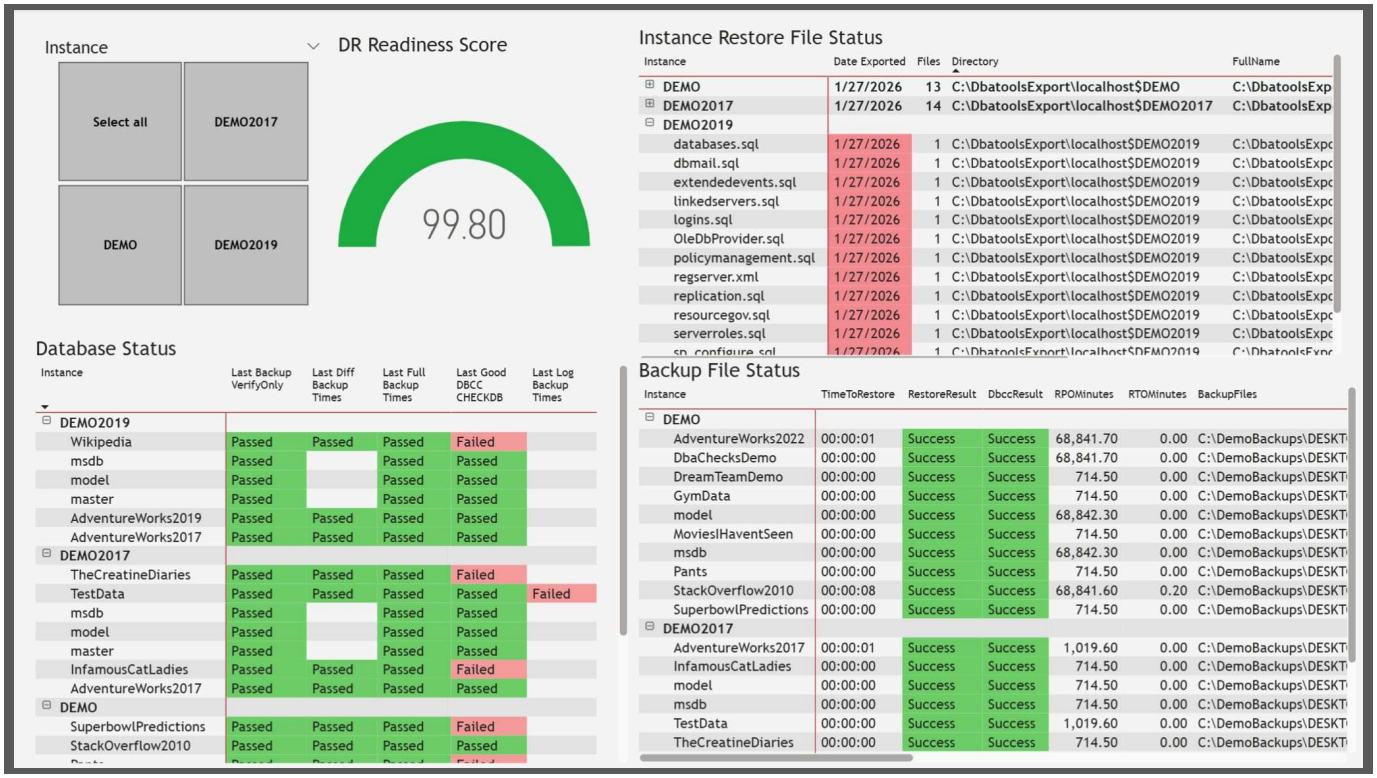
```
    We need to test our DR process! If we don't test our DR, we don't really have DR.

    Some extra fun pester checks you can use around your recovery testing by Chrissy Lemaire
        can be found at sqlps.io/doomsday
#>

# Test your recovery process
$files = Get-ChildItem -Path $restoreSqlInstancePath
$files | ForEach-Object {
    Write-Output "Restoring $PSItem"
    Invoke-DbaQuery -File $PSItem -SqlInstance $restoreSqlInstance -ErrorAction Ignore -Verbose
}
```

```
Restoring databases.sql
Restoring dbmail.sql
Restoring extendedevents.sql
Restoring logins.sql
VERBOSE: Changed database context to 'master'.
VERBOSE: Changed database context to 'master'.
VERBOSE: Cannot grant, deny, or revoke permissions to sa, dbo, entity owner, information_schema, sys, or yourself.
VERBOSE: Changed database context to 'InfamousCatLadies'.
VERBOSE: Changed database context to 'TestData'.
VERBOSE: Changed database context to 'TheCreatineDiaries'.
Restoring OleDbProvider.sql
```

```
#Start PBi Dashboard for dbachecks health check data
Start-DbcPowerBi -FromDatabase
```

```
[13:15:14][Start-DbcPowerBi] Launching the dbachecks dashboard. This may take a moment.
```





Announcing Free SQL Server Performance Monitoring

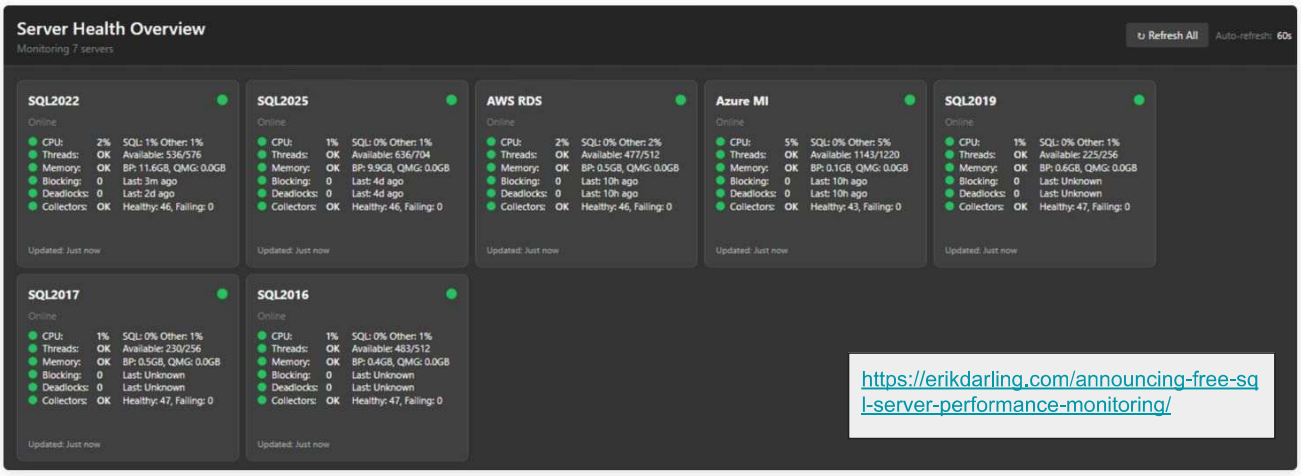
For years, enterprise SQL Server monitoring tools have been charging you an arm and a leg for stagnating products made by people who have never done an ounce of SQL Server performance tuning or DBA work. They phone home with your data, lock you into annual contracts, and still can't tell you why your server is slow without you doing half the work yourself.

That is, if you survive the installation process. Active Directory. Domain accounts. Firewalls. Security policies. Additional database servers. Annoying.

Or just send everything to the cloud, somewhere.

I got real tired of dealing with that, so I built a free, open-source SQL Server performance monitoring tool that isn't any of those things, *and* doesn't suck.

You can find it on GitHub.



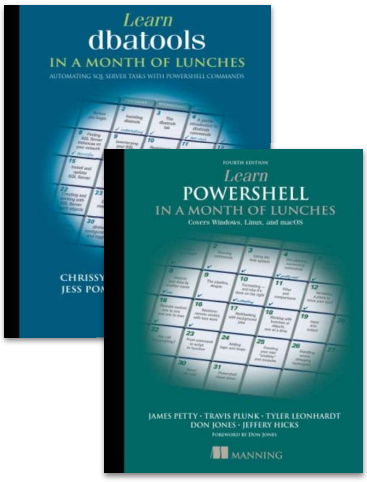
1… Liftoff!

Moonshot Thinking

Automation maturity ladder:

1. Manual
2. Script-assisted
3. Fully automated
4. Self-validating\healing

Recommended Additional Resources



Learn dbatools in a Month of Lunches:
https://www.manning.com/books/learn-dbatools-in-a-month-of-lunches

Learn PowerShell in a Month of Lunches:
https://www.manning.com/books/learn-powershell-in-a-month-of-lunches

dbachecks official site:
https://dbachecks.readthedocs.io/en/latest/

dbatools official site:
https://dbatools.io/getting-started/

Powershell/dbatools community on Slack:
https://dbatools.io/dbatools-is-now-on-the-sql-server-communitys-slack/

Thank you!

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